Data Flow Diagram (DFD)

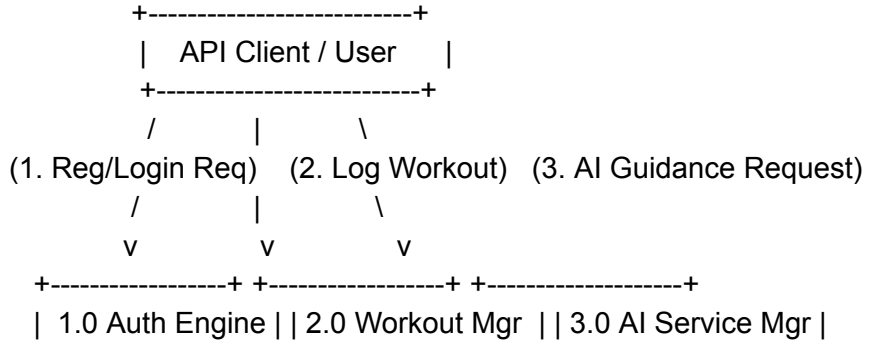
SMARTBRIDGE - Data Flow Diagram

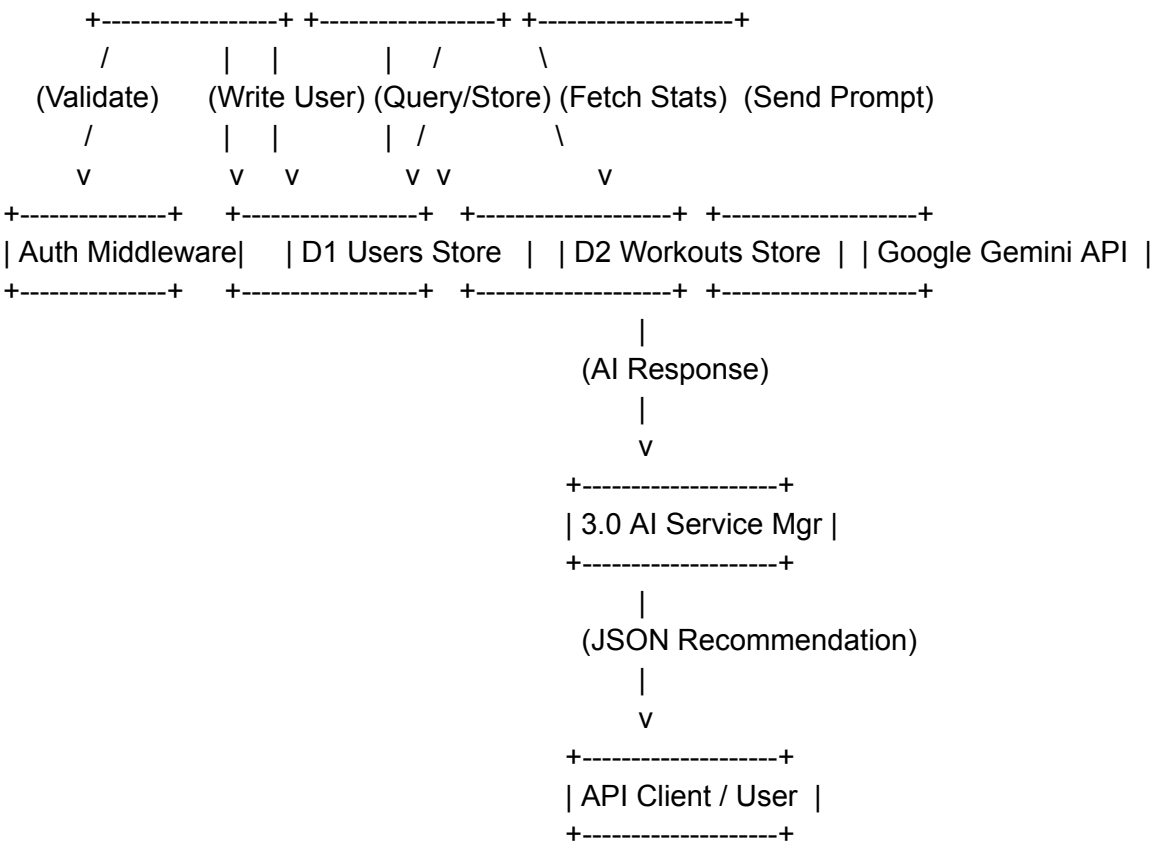
- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 2 Marks

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded	External Entity	An entity outside the system that sends or receives data (e.g., Client Application, Google Gemini API).
Rectangle (Numbered)	Process	An activity that transforms data (e.g., 1.0 User Auth, 2.0 Workout Manager, 3.0 Gemini AI Service).
Rectangle (Solid Header)	Data Store	A repository holding data for later use (e.g., D1 Users Collection, D2 Workouts Collection).
Labeled Arrow	Data Flow	Represents the movement of data between entities, processes, and stores.

DFD Data Flow Specifications





1. **Process 1.0 (User Authentication):** Receives credentials from **API Client**, hashes password using bcrypt, saves/verifies record in **D1: Users Store**, and returns JWT bearer token.
2. **Process 2.0 (Workout Management):** Authenticates JWT token via middleware, processes CRUD/search requests, and writes/queries records from **D2: Workouts Store**.
3. **Process 3.0 (Gemini AI Service Integration):** Receives user metrics (age, fitness goals, duration, calories), formats prompt, queries **Google Gemini API** (@google/genai), and returns structured JSON workout plans and fitness insights.